

PMBOK / PMI | PMB-08

Requirements Documentation and Traceability Matrix

What the solution must do, and the trace from need to verification

Meridian Group

Programme Helios — Infrastructure and Platform Stack for AI GPU Compute at Scale

Field	Detail
Document title	Requirements Documentation and Traceability Matrix
Document reference	PMB-08-HEL-T2-v1.0
Project / Programme	Helios Tranche 2 — Scale Build
Version	1.0
Status	Baselined
Date	18 Sep 2026
Author	D. Okonkwo, Programme Manager
Owner	Business analyst or project manager
Approver	Sponsor and named business owners
Classification	Internal
Retention	7 years from programme closure

Document control

Version history

Version	Date	Author	Summary of change
0.1	04 Jun 2026	D. Okonkwo, Programme Manager	Initial draft
0.2	21 Jul 2026	D. Okonkwo, Programme Manager	Updated following workstream lead review
1.0	18 Sep 2026	D. Okonkwo, Programme Manager	Baselined at the Tranche 2 funding gate

Reviewers

Name	Role	Review scope	Date reviewed
T. Nakamura, Chief Platform Architect	Chief Platform Architect	Technical content and solution alignment	08 Sep 2026
L. Haddad, CISO	CISO	Security, assurance and residency content	10 Sep 2026
C. Whitfield, Finance Business Partner	Finance Business Partner	Cost, funding and benefit figures	11 Sep 2026
M. Verity, Head of P3O	Head of P3O	Governance standards and completeness	15 Sep 2026

Approval

This product is baselined once approval is recorded below. Any subsequent change must go through change control.

Name	Role	Decision	Date
R. Al Mazrouei, Group CTO	Senior Responsible Owner / Group CTO	Approved	18 Sep 2026
H. Farrell, Director of Infrastructure	Director of Infrastructure	Approved	18 Sep 2026
L. Haddad, CISO	CISO	Approved subject to closure of ISS-003	18 Sep 2026

Related products

Reference	Product	Relationship
PMB-01	Project Charter	High-level requirements originate here
PMB-09	Project Scope Statement	Scope derives from requirements
PMB-17	Quality Management Plan	Verification methods

How to use this template

Aspect	Guidance
Purpose	Records the requirements the project must satisfy and traces each one from its business need through design and delivery to its verification, so that nothing is built without a reason and nothing required is missed.
Framework	PMBOK / PMI — Scope Management — Collect Requirements
Created when	During planning; elaborated progressively in iterative approaches
Reviewed / updated	At every phase gate and whenever requirements change
Owner	Business analyst or project manager
Approver	Sponsor and named business owners
Format options	Document, presentation, register entry, or tool record — the content matters, not the medium.
Tailoring	In an agile or hybrid delivery, the matrix traces epics and features rather than statements. Keep the trace; change the unit.

This is a worked example. Every field has been completed with illustrative content for Programme Helios, a fictional AI GPU infrastructure and platform programme. All example content is shown in grey italic — the black text is the template itself. Replace the grey italic content with your own.

Quality criteria

Use these to check the product is fit for purpose before submitting it for approval.

- Every requirement is traceable to a business objective or a regulatory obligation
- Requirements are testable — each has a stated verification method
- Non-functional requirements are captured, not only functional ones
- Priority uses a method that forces differentiation, such as MoSCoW
- Requirements are uniquely referenced and version-controlled

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1. Requirements elicitation summary

Technique	Participants	Date	Requirements produced
Written pack circulated 48 hours in advance	Programme Board	23 Apr 2027	The platform shall provide scheduled access to GPU capacity for at least three tenants with enforced fair-share quotas
Live walkthrough with a recorded session	Programme Manager and workstream leads	28 May 2027	The platform shall support both interactive notebook workloads and long-running distributed training jobs
Dashboard published to the programme site	CISO and Security Architect	25 Jun 2027	Training data classified as Restricted shall not leave UAE territory at rest or in transit
Workshop with the affected workstream leads	Business Owners and tenant representatives	18 Sep 2026	All container images shall be signed and admitted only from the approved internal registry
Automated report from the observability platform	Service Operations and FM provider	15 Oct 2026	The fabric shall sustain at least 92% all-reduce efficiency at 256-GPU job scale
Written pack circulated 48 hours in advance	Agile Release Train — all teams	30 Nov 2026	The storage tier shall sustain 1.2 TB/s aggregate read throughput to the compute estate

2. Requirements register

Ref	Requirement	Type	Source	Business objective	Priority	Acceptance criteria	Owner	Status
REQ-001	The platform shall provide scheduled access to GPU capacity for at least three tenants with enforced fair-share quotas	Functional / Non-functional / Regulatory / Transition	Business case	Improved utilisation	M/S/C/W	Three tenants running concurrently with quota enforcement demonstrated under contention	P. Raghavan, Platform Engineering Lead	Verified
REQ-002	The platform shall support both interactive notebook workloads and long-running distributed training jobs	Functional	AI Research	Faster training cycle time	M	A 64-GPU distributed job and 20 concurrent notebooks run without interference	N. Ozturk, ML Platform Lead	Verified

Ref	Requirement	Type	Source	Business objective	Priority	Acceptance criteria	Owner	Status
REQ-003	Training data classified as Restricted shall not leave UAE territory at rest or in transit	Regulatory	Group data residency policy v4.2	Residency assurance	M	Network egress controls and storage placement verified by independent test	L. Haddad, CISO	Verified
REQ-004	All container images shall be signed and admitted only from the approved internal registry	Non-functional	Group security standard	Supply chain integrity	M	Unsigned image admission is refused; attempt logged and alerted	K. Adeyemi, Security Architect	Verified
REQ-005	The fabric shall sustain at least 92% all-reduce efficiency at 256-GPU job scale	Non-functional	Architecture design	Training throughput	M	NCCL benchmark result at or above 92% on three consecutive runs	J. Ferreira, Network Lead	Failed
REQ-006	The storage tier shall sustain 1.2 TB/s aggregate read throughput to the compute estate	Non-functional	Architecture design	Training throughput	M	Benchmark against a representative dataset sustained for 30 minutes	I. Petrova, Data Platform Lead	Verified
REQ-007	Users shall be able to self-serve a working training environment within two hours of request	Functional	Business case	Time to provision	S	Median request-to-usable time under two hours across 20 sampled requests	P. Raghavan, Platform Engineering Lead	In build
REQ-008	The platform shall expose per-tenant consumption data suitable for chargeback	Functional	Finance	Cost transparency	M	Monthly consumption report reconciles to telemetry within 2%	C. Whitfield, Finance Business Partner	Verified
REQ-009	Vendor support access shall be brokered through a monitored in-country jump host with session recording	Non-functional	Group security standard	Least privilege	M	No direct vendor access path exists; sessions are recorded and retained 12 months	K. Adeyemi, Security Architect	Verified
REQ-010	GPU utilisation telemetry shall be collected at 10-second resolution and retained for 13 months	Non-functional	Benefits plan	Benefit measurement	S	Telemetry present at stated resolution and retention verified	G. Lindqvist, Service Operations Lead	Verified
REQ-	The platform shall recover	Non-	Service	Resilience	S	Failover exercise completed	G. Lindqvist,	In build

Ref	Requirement	Type	Source	Business objective	Priority	Acceptance criteria	Owner	Status
011	<i>scheduled workloads within four hours of a single hall failure</i>	<i>functional</i>	<i>design</i>			<i>within the stated RTO</i>	<i>Service Operations Lead</i>	

3. Traceability matrix

Req ref	Business objective	Deliverable / WBS ref	Design ref	Test case ref	Verified date	Accepted by	Status
REQ-001	<i>Improved utilisation</i>	REQ-001	REQ-001	REQ-001	26 Mar 2027	V. Chandra	Verified
REQ-002	<i>Faster training cycle time</i>	REQ-002	REQ-002	REQ-002	26 Mar 2027	Dr S. Iyer	Verified
REQ-003	<i>Residency assurance</i>	REQ-003	REQ-003	REQ-003	14 Apr 2027	L. Haddad	Verified
REQ-004	<i>Supply chain integrity</i>	REQ-004	REQ-004	REQ-004	14 Apr 2027	L. Haddad	Verified
REQ-005	<i>Training throughput</i>	REQ-005	REQ-005	REQ-005	02 Oct 2026	J. Ferreira	Failed
REQ-006	<i>Training throughput</i>	REQ-006	REQ-006	REQ-006	20 Jan 2027	I. Petrova	Verified
REQ-007	<i>Time to provision</i>	REQ-007	REQ-007	REQ-007	Not yet	Pending	In build
REQ-008	<i>Cost transparency</i>	REQ-008	REQ-008	REQ-008	31 Mar 2027	C. Whitfield	Verified
REQ-009	<i>Least privilege</i>	REQ-009	REQ-009	REQ-009	14 Apr 2027	L. Haddad	Verified
REQ-010	<i>Benefit measurement</i>	REQ-010	REQ-010	REQ-010	31 Jan 2027	G. Lindqvist	Verified

4. Non-functional requirements

Security, performance, availability, accessibility, data protection and regulatory requirements. These are the ones most often discovered late.

Ref	Category	Requirement	Measure	Verification method	Owner
NFR-01	Security	All container images signed and admitted only from the internal registry	Zero unsigned admissions	Negative admission test, quarterly	K. Adeyemi, Security Architect
NFR-02	Security	No standing vendor privilege; all access brokered and recorded	Zero standing vendor accounts	Access review, monthly	L. Haddad, CISO
NFR-03	Data residency	Restricted data remains within UAE borders at rest and in transit	Zero egress events for Restricted data	Egress control test and log review	L. Haddad, CISO
NFR-04	Performance	All-reduce efficiency at or above 92% at 256-GPU scale	92%	NCCL benchmark, three consecutive runs	J. Ferreira, Network Lead
NFR-05	Performance	Storage sustains 1.2 TB/s aggregate read	1.2 TB/s for 30 minutes	Workload-representative benchmark	I. Petrova, Data Platform Lead
NFR-06	Availability	Scheduled workloads recover within four hours of a single hall failure	4-hour RTO	Failover exercise	G. Lindqvist, Service Operations Lead
NFR-07	Safety	Coolant leak alerts the NOC within 60 seconds and isolates the rack	60 seconds	Simulated leak drill	A. Kowalski, Data Centre Lead
NFR-08	Sustainability	Rolling 12-month PUE of 1.25 or better when fully loaded	1.25	BMS metering export	A. Kowalski, Data Centre Lead

5. Requirements change log

Date	Req ref	Change	Reason	Change request ref	Approved by
31 Jan 2027	REQ-001	Add a second CDU to hall A to reach the 60 kW design point	Supply chain provenance for AI tooling is weak; admission control is the only enforceable point	Add coolant leak detection integration to the NOC	T. Nakamura, Chief Platform Architect
26 Feb 2027	REQ-002	Extend contractor SRE cover by six months	Grid capacity at DXB-1 supports Tranche 2 in full, so a second site is	Reduce Tranche 2 GPU count from 2,048	F. Mansour, Business Change Lead

Date	Req ref	Change	Reason	Change request ref	Approved by
			not yet justified	to 1,792	
31 Mar 2027	REQ-003	Defer confidential computing enabler from Tranche 2 to Tranche 3	Physical lead times do not tolerate emerging requirements, so this element is frozen before order	Add a second CDU to hall A to reach the 60 kW design point	D. Okonkwo, Programme Manager
23 Apr 2027	REQ-004	Integration partner claim for additional structural works	Consolidation is the primary source of the benefit case; utilisation across siloed estates averages 31%	Extend contractor SRE cover by six months	K. Adeyemi, Security Architect
28 May 2027	REQ-005	Bridge the hall B row 3 service void rather than relocate racks	Air cooling would cap density at 12 kW per rack and require 2.4 times the floor area	Defer confidential computing enabler from Tranche 2 to Tranche 3	I. Petrova, Data Platform Lead
25 Jun 2027	REQ-006	Dual-source 20% of Tranche 2 accelerator volume	Two independent schedulers fragment the GPU pool and reduce effective capacity	Integration partner claim for additional structural works	J. Ferreira, Network Lead
18 Sep 2026	REQ-007	Add coolant leak detection integration to the NOC	Supply chain provenance for AI tooling is weak; admission control is the only enforceable point	Bridge the hall B row 3 service void rather than relocate racks	V. Chandra, Head of AI Enablement

6. Rejected and deferred requirements

Record these. Requirements that vanish without a trace resurface as change requests.

Ref	Requirement	Rejected or deferred	Rationale	Decided by	Date
REQ-001	The platform shall provide scheduled access to GPU capacity for at least three tenants with enforced fair-share quotas	Enforced automatically in the pipeline; manual assertion is not accepted as evidence	Physical lead times do not tolerate emerging requirements, so this element is frozen before order	M. Verity, Head of P30	15 Dec 2026

Ref	Requirement	Rejected or deferred	Rationale	Decided by	Date
REQ-002	The platform shall support both interactive notebook workloads and long-running distributed training jobs	Reviewed at each end-of-tranche review and updated by change control only	Consolidation is the primary source of the benefit case; utilisation across siloed estates averages 31%	G. Lindqvist, Service Operations Lead	31 Jan 2027
REQ-003	Training data classified as Restricted shall not leave UAE territory at rest or in transit	Enforced automatically in the pipeline; manual assertion is not accepted as evidence	Air cooling would cap density at 12 kW per rack and require 2.4 times the floor area	H. Farrell, Director of Infrastructure	26 Feb 2027
REQ-004	All container images shall be signed and admitted only from the approved internal registry	Any deviation requires a recorded waiver accepted by the CISO with a stated expiry	Two independent schedulers fragment the GPU pool and reduce effective capacity	N. Ozturk, ML Platform Lead	31 Mar 2027
REQ-005	The fabric shall sustain at least 92% all-reduce efficiency at 256-GPU job scale	Applies to all workloads on the DXB-1 estate without exception	Air cooling would cap density at 12 kW per rack and require 2.4 times the floor area	B. Oyelaran, Quality and Assurance Lead	23 Apr 2027
REQ-006	The storage tier shall sustain 1.2 TB/s aggregate read throughput to the compute estate	Applies to all workloads on the DXB-1 estate without exception	Consolidation is the primary source of the benefit case; utilisation across siloed estates averages 31%	T. Nakamura, Chief Platform Architect	28 May 2027